

Random generalized Lotka-Volterra model on sparse graphs: Gaussian expansion of dynamic cavity method

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Abstract

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. INTRODUCTION

Intro on the subject.

II. METHOD

A. Random generalized Lotka-Volterra model on sparse graphs

We consider a generalized Lotka-Volterra (gLV) model describing the dynamics of an interacting community of N species, labeled by $i = 1, \dots, N$. The species interacts on an undirected random graph G with symmetric adjacency matrix $A = \{a_{ij}\}$, meaning that species i and j are interacting if $a_{ij} = a_{ji} = 1$, otherwise the matrix elements are zero. The time-dependent number density of individuals of species i is denoted by $x_i(t)$, and evolves in time according to

$$\frac{d}{dt}x_i(t) = x_i(t) \left[1 - x_i(t) + \sum_{j \neq i} a_{ij} J_{ij} x_j(t) \right] + \eta_i(t) + \lambda, \quad (1)$$

where we included, for the sake of generality, a Gaussian noise $\eta_i(t)$ with vanishing mean and variance $\langle \eta_i(t) \eta_{i'}(t') \rangle = 2Tg(x_i(t))\delta_{ii'}\delta(t-t')$, and a constant immigration rate λ . We will consider in what follows three choices for the function $g(x)$: (i) $g(x) = 0$ for deterministic gLV model, (ii) $g(x) = x$ for demographic noise, and (iii) $g(x) = x^2$ for environmental noise.

The couplings J_{ij} represent the (per capita) effect of species on one another. A negative coefficient J_{ij} indicates a competitive effect of species j on species i .

We consider here Gaussian J_{ij} with average $\overline{J_{ij}} = \overline{J_{ji}} = m/K$, variance $\text{Var}(J_{ij}) = \text{Var}(J_{ji}) = \sigma^2/K$ and covariance $\text{Corr}(J_{ij}, J_{ji}) = \gamma$. The overbar $\overline{\cdot}$ represents the average over the random couplings ensemble. The scaling of the moments with the average degree of the graph K is needed to produce a well-defined fully connected limit $K \rightarrow \infty$. The parameter $-1 \leq \gamma \leq 1$, being related to the correlation between the interaction coefficients J_{ij} and J_{ji} , measures the fraction of predator-prey pairs. Such a predator-prey pair consists of two species i and j for which J_{ij} and J_{ji} have opposite signs. If for example J_{ij} is positive and J_{ji} is negative, the presence of species i has a detrimental effect on species j , while the presence of species j is beneficial for species i . For any pair $i < j$ we can write

$$J_{ij} = \frac{m}{K} + \frac{\sigma}{\sqrt{K}} z_{ij}, \quad J_{ji} = \frac{m}{K} + \frac{\sigma}{\sqrt{K}} z_{ji}, \quad (2)$$

where z_{ij} and z_{ji} are Gaussian random variables with $\overline{z_{ij}} = 0$, $\overline{z_{ij}^2} = 1$, $\overline{z_{ij} z_{ji}} = \gamma$.

For $\gamma = 1$ we have $z_{ij} = z_{ji}$ with probability one, for $\gamma = 0$ z_{ij} and z_{ji} are uncorrelated, and for $\gamma = -1$ we have $z_{ij} = -z_{ji}$ with probability one. The latter, $\gamma = -1$, corresponds to having all the interactions of the predator-prey type. Increasing the value of γ this fraction decrease, until the opposite case, $\gamma = 1$ of no predator-prey type interactions. The other interactions can be of a mutualistic type (J_{ij} and J_{ji} both positive) or strictly competitive (J_{ij} and J_{ji} both negative).

37 B. Small coupling expansion of dynamic cavity equations

38 We use a small coupling expansion of the dynamic cavity equations in a way similar to [1] to derive an effective
 39 Langevin equation for non-interactive species. This approach begins by constructing the Martin-Siggia-Rose-Janssen-
 40 De Dominicis (MSRJD) dynamic partition function through path integrals of the trajectories [2–4]

$$Z = \left\langle \int \mathcal{D}\vec{x} \prod_i p_i(x_i(0)) \delta \left(\dot{x}_i(t) - x_i(t) \left[1 - x_i(t) + \sum_{j \neq i} a_{ij} J_{ij} x_j(t) + h_i(t) \right] - \eta_i(t) - \lambda \right) \right\rangle_{\vec{\eta}}, \quad (3)$$

41 where $\int \mathcal{D}\vec{x} = \prod_i \int \mathcal{D}x_i$ is a functional integral over the trajectories of the system, and $h_i(t)$ is an external field,
 42 needed to compute response functions. Following the steps of [1] we put forward a dynamic cavity Ansatz and expand
 43 the cavity marginals for small couplings. This results in a set of effective de-coupled Langevin equations on the cavity
 44 graph. For any edge (i, j) on the graph G the effective cavity population size $x_{i \setminus j}(t)$ evolves under the following
 45 stochastic process

$$\begin{aligned} \frac{d}{dt} x_{i \setminus j}(t) = & x_{i \setminus j}(t) \left[1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i \setminus j} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_{i \setminus j}(t') + h_{i \setminus j}(t) + \zeta_{i \setminus j}(t) \right] \\ & + \sqrt{2Tg(x_{i \setminus j}(t))} \xi_{i \setminus j}(t) + \lambda \end{aligned} \quad (4)$$

46 where $h_{i \setminus j}(t)$ is an external field, needed to compute response functions, and $\zeta_{i \setminus j}$ and $\xi_{i \setminus j}$ are Gaussian noises with
 47 vanishing mean and correlations

$$\langle \zeta_{i \setminus j}(t) \zeta_{i \setminus j}(t') \rangle_{i \setminus j} = \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t'), \quad \langle \xi_{i \setminus j}(t) \xi_{i \setminus j}(t') \rangle_{i \setminus j} = \delta(t - t'). \quad (5)$$

48 The terms $\mu_{k \setminus i}(t)$, $R_{k \setminus i}(t, t')$, and $C_{k \setminus i}(t, t')$ are respectively the cavity mean abundance, the cavity response
 49 function, and the cavity (connected) two-point correlation function. They are obtained self-consistently from the
 50 effective dynamics of the neighbors

$$\mu_{k \setminus i}(t) = \langle x_{k \setminus i}(t) \rangle_{k \setminus i}, \quad R_{k \setminus i}(t, t') = \frac{\delta}{\delta h_{k \setminus i}(t')} \langle x_{k \setminus i}(t) \rangle_{k \setminus i}, \quad C(t, t') = \langle x_{k \setminus i}(t) x_{k \setminus i}(t') \rangle_{k \setminus i} - \mu_{k \setminus i}(t) \mu_{k \setminus i}(t'), \quad (6)$$

51 where $\langle \dots \rangle_{k \setminus i}$ describes the average over the distribution of the effective process Eq. (4) for the edge (k, i) . The
 52 effective process is therefore de-coupled, and the interactions are captured by the self-consistent external fields $\mu_{k \setminus i}(t)$,
 53 $R_{k \setminus i}(t, t')$, and $C_{k \setminus i}(t, t')$. A detailed derivation can be found in Appendix A.

54 Once the cavity averages have been determined through self-consistency, one can obtain the effective de-coupled
 55 Langevin equations for the species population

$$\frac{d}{dt} x_i(t) = x_i(t) \left[1 - x_i(t) + \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_i(t') + h_i(t) + \zeta_i(t) \right] + \sqrt{2Tg(x_i(t))} \xi_i(t) + \lambda, \quad (7)$$

56 with Gaussian noises with vanishing mean and correlations

$$\langle \zeta_i(t) \zeta_i(t') \rangle_i = \sum_{k \in \partial i} J_{ik}^2 C_{k \setminus i}(t, t'), \quad \langle \xi_i(t) \xi_i(t') \rangle_i = \delta(t - t'). \quad (8)$$

57 The average $\langle \dots \rangle_i$ is performed over the effective de-coupled process Eq. (7).

C. Numerical integration

We propose a numerical integration scheme, mostly based on what has been proposed in [5]. The algorithm iteratively updates the self-consistent external fields until convergence.

We start with initial guesses for the cavity mean abundances $\{\mu_{i \setminus j}\}_{(i,j) \in E}$, the cavity correlations $\{C_{i \setminus j}\}_{(i,j) \in E}$, and the cavity responses $\{R_{i \setminus j}\}_{(i,j) \in E}$ for each edge of the graph. At iteration n , for each edge (i, j) the following steps are followed:

1. We sample N_{traj} Gaussian paths $(\zeta_{i \setminus j}(t), \xi_{i \setminus j}(t))$, where $\zeta_{i \setminus j}(t)$ and $\xi_{i \setminus j}(t)$ have, respectively, covariance matrices $C_{i \setminus j}^\zeta(t, t') = \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t')$ and $C_{i \setminus j}^\xi(t, t') = \delta(t, t')$.
2. For each path, we numerically integrate the effective cavity equation with an initial condition sampled according to $p_0(x)$:

$$\begin{aligned} \frac{d}{dt} x_{i \setminus j}(t) &= x_{i \setminus j}(t) \left[1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i \setminus j} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_{i \setminus j}(t') + \zeta_{i \setminus j}(t) \right] \\ &\quad + \sqrt{2Tg(x_{i \setminus j}(t))} \xi_{i \setminus j}(t) + \lambda, \\ x_{i \setminus j}(0) &\sim p_0(x_{i \setminus j}(0)). \end{aligned}$$

3. We compute the new values of the cavity self-consistent fields as averages over the paths:

$$\begin{aligned} \mu_{i \setminus j}^{\text{new}}(t) &= \mathbb{E}_{\text{paths}} [x_{i \setminus j}(t)], \\ C_{i \setminus j}^{\text{new}}(t, t') &= \mathbb{E}_{\text{paths}} [x_{i \setminus j}(t) x_{i \setminus j}(t')] - \mu_{i \setminus j}^{\text{new}}(t) \mu_{i \setminus j}^{\text{new}}(t'), \\ R_{i \setminus j}^{\text{new}}(t, t') &= \mathbb{E}_{\text{paths}} [R_{i \setminus j}^{\text{sp}}(t, t')]. \end{aligned}$$

4. We update softly the cavity self-consistent fields: $X_{i \setminus j}^{\text{updated}} = (1 - d)X_{i \setminus j}^{\text{new}} + dX_{i \setminus j}$, where X is μ , C and R . d is the damping factor.

The terms $R_{i \setminus j}^{\text{sp}}(t, t')$ are single-path response functions, and can be computed through numerical integration of the equation obtained by directly applying $\delta/\delta h_{i \setminus j}(t')$ to Eq. (4). More details can be found in Appendix B.

The algorithm proceeds in iterations, where each iteration consists of updating all edges (i, j) of the graph. These iterations repeat until convergence is achieved, defined as the change in the cavity self-consistent fields between successive iterations dropping below a threshold ε . For computational efficiency, we evaluate convergence based on the stabilization of correlation functions only.

$$\max_{(i,j)} \|C_{i \setminus j}^{\text{updated}} - C_{i \setminus j}^{\text{new}}\| < \epsilon, \quad (9)$$

where $\|\cdot\|$ is some norm to be defined. When convergence is reached, we can compute the local effective process by using the self-consistent cavity fields in Eq. (7). In practice, for each node i of the graph we sample N_{traj} Gaussian paths $(\zeta_i(t), \xi_i(t))$, where $\zeta_i(t)$ and $\xi_i(t)$ have, respectively, covariance matrices $C_i^\zeta(t, t') = \sum_{k \in \partial i} J_{ik}^2 C_{k \setminus i}(t, t')$ and $C_i^\xi(t, t') = \delta(t, t')$. We then numerically integrate the effective marginal equation with an initial condition sampled according to $p_0(x)$:

$$\begin{aligned} \frac{d}{dt} x_i(t) &= x_i(t) \left[1 - x_i(t) + \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_i(t') + \zeta_i(t) \right] \\ &\quad + \sqrt{2Tg(x_i(t))} \xi_i(t) + \lambda, \\ x_i(0) &\sim p_0(x_i(0)). \end{aligned}$$

The full average quantities are then obtained by averaging over the sampled paths:

$$\begin{aligned} \mu_i(t) &= \mathbb{E}_{\text{paths}} [x_i(t)], \\ C_i(t, t') &= \mathbb{E}_{\text{paths}} [x_i(t) x_i(t')] - \mu_i(t) \mu_i(t'), \\ R_i(t, t') &= \mathbb{E}_{\text{paths}} [R_i^{\text{sp}}(t, t')]. \end{aligned}$$

More details on the numerical integration can be found in Appendix C.

84 D. Gaussian closure approximation

85 While a direct numerical integration of the effective process is possible, we can also use a Gaussian closure
 86 approximation to obtain analytical results. This approach is based on the assumption that the distribution of the
 87 population sizes $x_i(t)$ is Gaussian, which allows us to compute the moments of the process analytically. The Gaussian
 88 closure approximation leads to a set of self-consistent equations for the first and second moments of the population
 89 sizes, which can be solved iteratively.

90 In particular, by taking averages of the effective cavity process Eq. (4) and approximating three-times averages as
 91 sums of products of one- and two-times averages, we obtain the following self-consistent equations for the first and
 92 second moments of the population sizes:

$$\begin{aligned} \frac{d}{dt}\mu_{i\setminus j}(t) &= \langle \dot{x}_{i\setminus j}(t) \rangle_{i\setminus j} = \mu_{i\setminus j}(t) - C_{i\setminus j}(t, t) - \mu_{i\setminus j}^2(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) \mu_{i\setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_{i\setminus j}(t, s) + \mu_{i\setminus j}(t) \mu_{i\setminus j}(s)) + \lambda, \end{aligned} \quad (10)$$

$$\begin{aligned} \frac{\partial}{\partial t} C_{i\setminus j}(t, t') &= \langle \dot{x}_{i\setminus j}(t) x_{i\setminus j}(t') \rangle_{i\setminus j} = (1 - 2\mu_{i\setminus j}(t)) C_{i\setminus j}(t, t') + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) C_{i\setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (\mu_{i\setminus j}(t) C_{i\setminus j}(s, t') + C_{i\setminus j}(t, t') \mu_{i\setminus j}(s)) + 2T \mathcal{L}_{i\setminus j}(t) R_{i\setminus j}(t', t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}^2 \left(\int_0^{t'} ds C_{k\setminus i}(t, s) \mu_{i\setminus j}(t) R_{i\setminus j}(t', s) + \int_0^t ds C_{k\setminus i}(t, s) R_{i\setminus j}(t, s) \mu_{i\setminus j}(t') \right), \end{aligned} \quad (11)$$

$$\begin{aligned} \frac{\partial}{\partial t} R_{i\setminus j}(t, t') &= \langle \dot{x}_{i\setminus j}(t) \hat{x}_{i\setminus j}(t') \rangle_{i\setminus j} = (1 - 2\mu_{i\setminus j}(t)) R_{i\setminus j}(t, t') + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) R_{i\setminus j}(t, t') + \delta(t - t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \left(\int_0^t ds R_{k\setminus i}(t, s) R_{i\setminus j}(t, t') \mu_{i\setminus j}(s) + \int_{t'}^t ds \mu_{i\setminus j}(t) R_{k\setminus i}(t, s) R_{i\setminus j}(s, t') \right), \end{aligned} \quad (12)$$

93 where we have introduced the function

$$\mathcal{L}_{i\setminus j}(t) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1 & g(x) = x & \text{demographic} \\ \mu_{i\setminus j}(t) & g(x) = x^2 & \text{environmental.} \end{cases} \quad (13)$$

94 An equivalent set of equations can be obtained for the full marginal process:

$$\begin{aligned} \frac{d}{dt}\mu_i(t) &= \langle \dot{x}_i(t) \rangle_i = \mu_i(t) - C_i(t, t) - \mu_i^2(t) + \sum_{k \in \partial i} J_{ik} \mu_{k\setminus i}(t) \mu_i(t) \\ &+ \sum_{k \in \partial i} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_i(t, s) + \mu_i(t) \mu_i(s)) + \lambda, \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{\partial}{\partial t} C_i(t, t') &= \langle \dot{x}_i(t) x_i(t') \rangle_i = (1 - 2\mu_i(t)) C_i(t, t') + \sum_{k \in \partial i} J_{ik} \mu_{k\setminus i}(t) C_i(t, t') \\ &+ \sum_{k \in \partial i} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (\mu_i(t) C_i(s, t') + C_i(t, t') \mu_i(s)) + 2T \mathcal{L}_i(t) R_i(t', t) \\ &+ \sum_{k \in \partial i} J_{ik}^2 \left(\int_0^{t'} ds C_{k\setminus i}(t, s) \mu_i(t) R_i(t', s) + \int_0^t ds C_{k\setminus i}(t, s) R_i(t, s) \mu_i(t') \right), \end{aligned} \quad (15)$$

$$\begin{aligned} \frac{\partial}{\partial t} R_i(t, t') &= \langle \dot{x}_i(t) \hat{x}_i(t') \rangle_i = (1 - 2\mu_i(t)) R_i(t, t') + \sum_{k \in \partial i} J_{ik} \mu_{k\setminus i}(t) R_i(t, t') + \delta(t - t') \\ &+ \sum_{k \in \partial i} J_{ik} J_{ki} \left(\int_0^t ds R_{k\setminus i}(t, s) R_i(t, t') \mu_i(s) + \int_{t'}^t ds \mu_i(t) R_{k\setminus i}(t, s) R_i(s, t') \right), \end{aligned} \quad (16)$$

95 where the function $\mathcal{L}_i(t)$ is defined as

$$\mathcal{L}_i(t) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1 & g(x) = x & \text{demographic} \\ \mu_i(t) & g(x) = x^2 & \text{environmental.} \end{cases} \quad (17)$$

96 1. Stationary state

97 Assuming the system reaches a stationary state, we can set $\mu(t) = \mu$, $C(t, t') = C_i(t - t')$, and $R(t, t') = R(t - t')$.
 98 This corresponds to assuming the system to be in the single equilibrium state. The self-consistent cavity equations
 99 then simplify to

$$\begin{aligned} \frac{d}{dt}\mu_{i \setminus j} = 0 &= \mu_{i \setminus j} \left(1 - \mu_{i \setminus j} + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i} + \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \chi_{k \setminus i} \mu_{i \setminus j} \right) - C_{i \setminus j}(0) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^\infty du R_{k \setminus i}(u) C_{i \setminus j}(u) + \lambda, \end{aligned} \quad (18)$$

$$\begin{aligned} \frac{d}{d\tau} C_{i \setminus j}(\tau) &= C_{i \setminus j}(\tau) \left(1 - 2\mu_{i \setminus j} + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i} + \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \chi_{k \setminus i} \mu_{i \setminus j} \right) + 2T \mathcal{L}_{i \setminus j} R_{i \setminus j}(-\tau) \\ &+ \mu_{i \setminus j} \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^\infty du R_{k \setminus i}(u) C_{i \setminus j}(\tau - u) \\ &+ \mu_{i \setminus j} \sum_{k \in \partial i \setminus j} J_{ik}^2 \int_0^\infty du (C_{k \setminus i}(u - \tau) R_{i \setminus j}(u) + C_{k \setminus i}(u) R_{i \setminus j}(u)), \end{aligned} \quad (19)$$

$$\begin{aligned} \frac{d}{d\tau} R_{i \setminus j}(\tau) &= R_{i \setminus j}(\tau) \left(1 - 2\mu_{i \setminus j} + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i} + \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \chi_{k \setminus i} \mu_{i \setminus j} \right) + \delta(\tau) \\ &+ \mu_{i \setminus j} \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^\tau du R_{k \setminus i}(u) R_{i \setminus j}(\tau - u), \end{aligned} \quad (20)$$

100 where we have introduced the time shift $\tau = t - t'$, and the susceptibility $\chi_{i \setminus j} = \int_0^\infty du R_{i \setminus j}(u)$, which is the integrated
 101 response of the population i in the cavity graph $\partial i \setminus j$ to a local perturbation. We have defined the function $\mathcal{L}_{i \setminus j}$ as

$$\mathcal{L}_{i \setminus j} = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1 & g(x) = x & \text{demographic} \\ \mu_{i \setminus j} & g(x) = x^2 & \text{environmental.} \end{cases} \quad (21)$$

102 2. Fixed Point + Laplace

103 I have the intuition that we have some chances of going ahead exploiting Laplace as in the linear case for C and R ,
 104 but having a fixed point in μ .

105 Assuming that μ reached the fixed point it is easy to show that:

$$\begin{aligned} p \hat{R}_{i \setminus j}(p) + \hat{R}_{i \setminus j}(0) &= \left[1 - 2\mu_{i \setminus j} + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i} \right] \hat{R}_{i \setminus j}(p) + \\ &+ \mu_{i \setminus j} \hat{R}_{i \setminus j}(p) \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} (\chi_{k \setminus i} + \hat{R}_{k \setminus i}(p)) \end{aligned} \quad (22)$$

106 Similarly:

$$\begin{aligned}
p\hat{C}_{i\setminus j}(p) + \hat{C}_{i\setminus j}(0) &= \left[1 - 2\mu_{i\setminus j} + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}\right] \hat{C}_{i\setminus j}(p) + \\
&+ \mu_{i\setminus j} \hat{C}_{i\setminus j}(p) \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}(\chi_{k\setminus i} + \hat{R}_{k\setminus i}(p)) + \\
&+ \mu_{i\setminus j} \hat{R}_{i\setminus j}(p) \sum_{k \in \partial i \setminus j} J_{ik}^2 \hat{C}_{k\setminus i}(p) + \mu_{i\setminus j} \chi_{i\setminus j} \sum_{k \in \partial i \setminus j} J_{ik}^2 \hat{C}_{k\setminus i}(p)
\end{aligned} \tag{23}$$

107 where the last term in this expression is *dirty*. I assumed that: $\mathcal{L}\left[\int_0^t du C_{k\setminus i}(u) R_{k\setminus i}(u)\right] = \chi_{i\setminus j} \hat{C}_{k\setminus i}(p)$.

108 Finally, for the fixed point equation:

$$0 = \mu_{i\setminus j}(1 + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}) - \mu_{i\setminus j}^2(1 - \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}) - q_{i\setminus j}(1 - \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}) \tag{24}$$

109 that looks prettier this way:

$$0 = \mu_{i\setminus j}^2 - \mu_{i\setminus j} \frac{1 + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}}{1 - \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}} + q_{i\setminus j} \tag{25}$$

110 One can try a bit more and look for fixed point equations also for C . I get for q :

$$0 = \left[1 - 2\mu_{i\setminus j} + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i} + 2\mu_{i\setminus j} \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}\right] q_{i\setminus j} + \mu_{i\setminus j} \chi_{i\setminus j} \sum_{k \in \partial i \setminus j} J_{ik}^2 q_{k\setminus i} \tag{26}$$

111 E. Fixed-point equations for the deterministic model

112 We focus on the deterministic model without immigration, where $g(x) = 0$ and $\lambda = 0$. In this case, for certain
113 values of the parameters m , σ and γ , we can assume that the system reaches a steady state which does not depend on
114 the initial condition. We further assume that the dynamics reaches a unique fixed-point $\lim_{t \rightarrow \infty} x_{i\setminus j}(t) = x_{i\setminus j}^*$, and
115 therefore $\lim_{t \rightarrow \infty} \zeta_{i\setminus j}(t) = \zeta_{i\setminus j}^*$. The fixed point values $x_{i\setminus j}^*$, $\zeta_{i\setminus j}^*$ are both random variables. It follows that the mean
116 cavity abundance and the cavity correlation become time-independent variables $\lim_{t \rightarrow \infty} \mu_{i\setminus j}(t) = \mu_{i\setminus j}^* = \langle x_{i\setminus j}^* \rangle_{i\setminus j}^*$,
117 and $\lim_{t, t' \rightarrow \infty} C_{i\setminus j}(t, t') = q_{i\setminus j}^* = \langle (x_{i\setminus j}^*)^2 \rangle_{i\setminus j}^* - (\mu_{i\setminus j}^*)^2$. Within this assumption, the cavity response function is time-
118 translational-invariant, i.e., $R_{i\setminus j}(t, t')$ depends only on the time difference $\tau = t - t'$. Due to causality, $R_{i\setminus j}(\tau) = 0$
119 for $\tau \leq 0$, and we define the susceptibility as $\chi_{i\setminus j}^* = \int_0^\infty d\tau R_{i\setminus j}^*(\tau)$.

120 The fixed point satisfies the self-consistent equation

$$x_{i\setminus j}^* \left[1 - x_{i\setminus j}^* + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}^* + \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}^* x_{i\setminus j}^* + \xi_{i\setminus j}^* + h_{i\setminus j} \right] = 0, \tag{27}$$

121 where $h_{i\setminus j}$ is a constant external field needed to obtain the integrated cavity response, and the noise is Gaussian with
122 zero mean $\langle \xi_{i\setminus j}^* \rangle$ and variance $\langle (\xi_{i\setminus j}^*)^2 \rangle = \sum_{k \in \partial i \setminus j} J_{ik}^2 q_{k\setminus i}^* = \Sigma_{i\setminus j}^2$. We can therefore write $\xi_{i\setminus j}^* = -\Sigma_{i\setminus j} s$, where s is a
123 standard Gaussian variable with zero mean and unit variance. Eq. (27) admits non-vanishing solution

$$x_{i\setminus j}^*(s) = \frac{\Sigma_{i\setminus j}(\Delta_{i\setminus j} - s) + h_{i\setminus j}}{\Gamma_{i\setminus j}} \Theta\left(\frac{\Sigma_{i\setminus j}(\Delta_{i\setminus j} - s) + h_{i\setminus j}}{\Gamma_{i\setminus j}}\right), \tag{28}$$

124 where $\Theta(x)$ is the Heaviside step function, which ensures positivity of the fixed-point cavity abundance. We have
125 defined $\Delta_{i\setminus j} = (1 + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}^*)/\Sigma_{i\setminus j}$ and $\Gamma_{i\setminus j} = 1 - \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}^*$.

By averaging over the Gaussian distribution of s we obtain a set of self-consistent equations for the parameters

$$\mu_{i\setminus j}^* = \frac{\Sigma_{i\setminus j}}{\Gamma_{i\setminus j}} \left[\Theta(\Gamma_{i\setminus j}) \int_{-\infty}^{\Delta_{i\setminus j}} Ds (\Delta_{i\setminus j} - s) + \Theta(-\Gamma_{i\setminus j}) \int_{\Delta_{i\setminus j}}^{\infty} Ds (\Delta_{i\setminus j} - s) \right] \quad (29a)$$

$$q_{i\setminus j}^* = \frac{\Sigma_{i\setminus j}^2}{\Gamma_{i\setminus j}^2} \left[\Theta(\Gamma_{i\setminus j}) \int_{-\infty}^{\Delta_{i\setminus j}} Ds (\Delta_{i\setminus j} - s)^2 + \Theta(-\Gamma_{i\setminus j}) \int_{\Delta_{i\setminus j}}^{\infty} Ds (\Delta_{i\setminus j} - s)^2 \right] - \left(\mu_{i\setminus j}^* \right)^2 \quad (29b)$$

$$\chi_{i\setminus j}^* = \frac{1}{\Gamma_{i\setminus j}} \left[\Theta(\Gamma_{i\setminus j}) \int_{-\infty}^{\Delta_{i\setminus j}} Ds + \Theta(-\Gamma_{i\setminus j}) \int_{\Delta_{i\setminus j}}^{\infty} Ds \right]. \quad (29c)$$

We observe that the integrals can be written in terms of the error function $\text{erf}(x) = \int_0^x ds e^{-s^2} 2/\sqrt{\pi}$. Defining $\varphi(x) = e^{-x^2/2}/\sqrt{2\pi}$ the normal distribution function and $\Phi(x) = [1 + \text{erf}(x/\sqrt{2})]/2$ its cumulative distribution function we obtain

$$\mu_{i\setminus j}^* = \frac{\Sigma_{i\setminus j}}{\Gamma_{i\setminus j}} \left[\Theta(\Gamma_{i\setminus j}) (\Delta_{i\setminus j} \Phi(\Delta_{i\setminus j}) + \varphi(\Delta_{i\setminus j})) + \Theta(-\Gamma_{i\setminus j}) (\Delta_{i\setminus j} \Phi(-\Delta_{i\setminus j}) - \varphi(-\Delta_{i\setminus j})) \right] \quad (30a)$$

$$q_{i\setminus j}^* = \frac{\Sigma_{i\setminus j}^2}{\Gamma_{i\setminus j}^2} \left[\Theta(\Gamma_{i\setminus j}) \left(\left(1 + \Delta_{i\setminus j}^2 \right) \Phi(\Delta_{i\setminus j}) + \Delta_{i\setminus j} \varphi(\Delta_{i\setminus j}) \right) + \Theta(-\Gamma_{i\setminus j}) \left(\left(1 + \Delta_{i\setminus j}^2 \right) \Phi(-\Delta_{i\setminus j}) - \Delta_{i\setminus j} \varphi(-\Delta_{i\setminus j}) \right) \right] - \left(\mu_{i\setminus j}^* \right)^2 \quad (30b)$$

$$\chi_{i\setminus j}^* = \frac{1}{\Gamma_{i\setminus j}} \left[\Theta(\Gamma_{i\setminus j}) \Phi(\Delta_{i\setminus j}) + \Theta(-\Gamma_{i\setminus j}) \Phi(-\Delta_{i\setminus j}) \right]. \quad (30c)$$

We can solve these equations by means of a message-passing algorithm, together with the definitions of $\Delta_{i\setminus j}$, $\Sigma_{i\setminus j}$ and $\Gamma_{i\setminus j}$, which are given by

$$\Delta_{i\setminus j} = \frac{1 + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}^*}{\sqrt{\sum_{k \in \partial i \setminus j} J_{ik}^2 q_{k\setminus i}^*}} \quad (31a)$$

$$\Sigma_{i\setminus j}^2 = \sum_{k \in \partial i \setminus j} J_{ik}^2 q_{k\setminus i}^* \quad (31b)$$

$$\Gamma_{i\setminus j} = 1 - \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \chi_{k\setminus i}^*. \quad (31c)$$

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Appendix A: Derivation of the effective cavity process

The calculation is based on the principles of [1]. The dynamical partition function of the Lotka-Volterra system can be written as

$$Z = \left\langle \int \mathcal{D}\vec{x} \prod_i p_i(x_i(0)) \delta(\text{equation of motion}) \right\rangle_{\vec{\eta}}, \quad (\text{A1})$$

where $\int \mathcal{D}\vec{x} = \prod_i \int \mathcal{D}x$ is a product of functional integrals, $\langle \dots \rangle_{\vec{\eta}} = (\prod_i \int \mathcal{D}\eta_i) \dots$ is the average over the noise realizations, and $\delta(\dots)$ is a functional Dirac delta function which restrict the integral to the paths that satisfy the stochastic dynamics. Using the Fourier representation of the delta function $\delta(x) \propto \int \mathcal{D}\hat{x} e^{-i \int dt \hat{x}(t)x(t)}$ we obtain

$$Z = \left\langle \int \mathcal{D}\vec{x} \mathcal{D}\vec{\hat{x}} \prod_i p_i(x_i(0)) e^{\int dt (-i\hat{x}_i(t)) [\dot{x}_i(t) - x_i(t)(1 - x_i + \sum_{j \neq i} a_{ij} J_{ij} x_j + h_i(t)) - \eta_i(t) - \lambda]} \right\rangle_{\vec{\eta}}. \quad (\text{A2})$$

The integral over time run over a fixed time interval, but we omit the limits of integration, here and elsewhere, for the sake of simplifying the notation. We introduce an auxiliary variable $f_i = 1 - x_i - \sum_{j \neq i} a_{ij} J_{ij} x_j(t) + h_i(t)$, which can be inserted in the calculation through a Dirac delta function. Integrating over the Gaussian noise we obtain

$$Z = \int \mathcal{D}\vec{x} \mathcal{D}\vec{\hat{x}} \mathcal{D}\vec{f} \mathcal{D}\vec{\hat{f}} \prod_{i \in V} p_i(x_i(0)) e^{S_i(x_i, \hat{x}_i, f_i, \hat{f}_i)} \prod_{(i,j) \in E} e^{S_{ij}(x_i, \hat{x}_i, f_i, \hat{f}_i, x_j, \hat{x}_j, f_j, \hat{f}_j)}, \quad (\text{A3})$$

where we have defined the local and interactive actions as

$$S_i = \int dt \left\{ -i\hat{x}_i(t) (\dot{x}_i(t) - x_i(t)f_i(t) - \lambda) - \frac{1}{2} 2Tg(x_i(t))\hat{x}_i^2(t) - i\hat{f}_i(t) [f_i(t) - (1 - x_i(t) + h_i(t))] \right\} \quad (\text{A4})$$

$$S_{ij} = - \int dt \left(J_{ij} i\hat{f}_i(t)x_j(t) + J_{ji} i\hat{f}_j(t)x_i(t) \right) \quad (\text{A5})$$

We put forward the following *dynamic cavity equations* as an ansatz for describing linearly coupled stochastic dynamics on sparse networks

$$c_{i \setminus j}[x_i, \hat{x}_i, f_i, \hat{f}_i] = \frac{1}{Z_{i \setminus j}} e^{S_i} \prod_{k \in \partial i \setminus j} \int \mathcal{D}x_k \mathcal{D}\hat{x}_k \mathcal{D}f_k \mathcal{D}\hat{f}_k c_{k \setminus i}[x_k, \hat{x}_k, f_k, \hat{f}_k] e^{S_{ki}}, \quad (\text{A6})$$

The full dynamic cavity marginals can be obtained from the cavity ones as follows

$$c_i[x_i, \hat{x}_i, f_i, \hat{f}_i] = \frac{1}{Z_i} e^{S_i} \prod_{k \in \partial i} \int \mathcal{D}x_k \mathcal{D}\hat{x}_k \mathcal{D}f_k \mathcal{D}\hat{f}_k c_{k \setminus i}[x_k, \hat{x}_k, f_k, \hat{f}_k] e^{S_{ki}}. \quad (\text{A7})$$

Expanding the cavity marginals Eq. (A6) at second order in the coupling parameters J_{ij} we obtain an effective single node action for the cavity marginal

$$c_{i \setminus j}[x_i, \hat{x}_i, f_i, \hat{f}_i] \approx \frac{p_i(x_i(0))}{Z_{i \setminus j}} e^{S_{i \setminus j}^{\text{eff}}}, \quad (\text{A8})$$

where the effective cavity action is defined as

$$S_{i \setminus j}^{\text{eff}} = \int dt \left\{ -i\hat{x}_i(t) (\dot{x}_i(t) - x_i(t)f_i(t) - \lambda) - \frac{1}{2} 2Tg(x_i(t))\hat{x}_i^2(t) - i\hat{f}_i(t) [f_i(t) - (1 - x_i(t) + h_i(t))] \right. \\ \left. + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) - \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int dt' R_{k \setminus i}(t, t') x_i(t') \right\} + \int dt dt' i\hat{f}(t) \left(\sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t') \right) i\hat{f}(t') \quad (\text{A9})$$

where $\mu_{k \setminus i}(t)$, $R_{k \setminus i}(t, t')$, and $C_{k \setminus i}(t, t')$ are respectively the cavity mean abundance, the cavity response function, and the cavity (connected) two-point correlation function. They are obtained self-consistently from the effective dynamics of the neighbors

$$\mu_{k \setminus i}(t) = \langle x_k(t) \rangle_{k \setminus i}, \quad R_{k \setminus i}(t, t') = \left\langle x_k(t) \dot{f}_k(t') \right\rangle_{k \setminus i}, \quad C(t, t') = \langle x_k(t) x_k(t') \rangle_{k \setminus i} - \mu_{k \setminus i}(t) \mu_{k \setminus i}(t'), \quad (\text{A10})$$

where $\langle \dots \rangle_{k \setminus i} = \int \mathcal{D}x_k \mathcal{D}\hat{x}_k \mathcal{D}f_k \mathcal{D}\hat{f}_k c_{k \setminus i}[x_k, \hat{x}_k, f_k, \hat{f}_k] \dots$ describes the local average over the cavity marginal.

The effective cavity action can be identified with a sort of local effective single-edge action for the variables on site i when the interaction term with site j is removed from the underlying graph.

Performing backward the steps of the path-integral representation, we obtain an effective stochastic dynamics in terms of the stochastic differential equation on the cavity graph

$$\begin{aligned} \frac{d}{dt} x_{i \setminus j}(t) = x_{i \setminus j}(t) & \left[1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i \setminus j} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_{i \setminus j}(t') \right. \\ & \left. + h_{i \setminus j}(t) + \zeta_{i \setminus j}(t) \right] + \sqrt{2Tg(x_{i \setminus j})} \xi_{i \setminus j}(t) + \lambda, \end{aligned} \quad (\text{A11})$$

where $\zeta_{i \setminus j}$ and $\xi_{i \setminus j}$ are Gaussian noises with vanishing mean and correlations

$$\langle \zeta_{i \setminus j}(t) \zeta_{i \setminus j}(t') \rangle_{i \setminus j} = \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t'), \quad \langle \xi_{i \setminus j}(t) \xi_{i \setminus j}(t') \rangle_{i \setminus j} = \delta(t - t'). \quad (\text{A12})$$

The terms $\mu_{k \setminus i}(t)$, $R_{k \setminus i}(t, t')$, and $C_{k \setminus i}(t, t')$ have to be considered as external parameters, obtained self-consistently as averages over the effective dynamics of the neighbors. In a similar way, an effective process for the original graph can be obtained as

$$\begin{aligned} \frac{d}{dt} x_i(t) = x_i(t) & \left[1 - x_i(t) + \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_i(t') \right. \\ & \left. + h_i(t) + \zeta_i(t) \right] + \sqrt{2Tg(x_i(t))} \xi_i(t) + \lambda, \end{aligned} \quad (\text{A13})$$

with Gaussian noises with vanishing mean and correlations

$$\langle \zeta_i(t) \zeta_i(t') \rangle_i = \sum_{k \in \partial i} J_{ik}^2 C_{k \setminus i}(t, t'), \quad \langle \xi_i(t) \xi_i(t') \rangle_i = \delta(t - t'). \quad (\text{A14})$$

Appendix B: Single-path response function

The single-path response function $R_{i \setminus j}^{\text{sp}}(t, t') = \delta / \delta h_{i \setminus j}(t') x_{i \setminus j}(t)$ can be obtained by directly applying the functional derivative $\delta / \delta h_{i \setminus j}(t')$ to Eq. (4):

$$\begin{aligned} \frac{\partial}{\partial t} R_{i \setminus j}^{\text{sp}}(t, t') &= R_{i \setminus j}^{\text{sp}}(t, t') (1 - 2x_{i \setminus j}(t)) + R_{i \setminus j}^{\text{sp}}(t, t') \sqrt{\frac{2T}{g(x_{i \setminus j}(t))}} g'(x_{i \setminus j}(t)) \xi_{i \setminus j}(t) \\ &+ R_{i \setminus j}^{\text{sp}}(t, t') \left(\sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) + \zeta_{i \setminus j}(t) \right) + x_{i \setminus j}(t) \delta(t - t') \\ &+ R_{i \setminus j}^{\text{sp}}(t, t') \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k \setminus i}(t, s) x_{i \setminus j}(s) + x_{i \setminus j}(t) \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k \setminus i}(t, s) R_{i \setminus j}^{\text{sp}}(s, t'). \end{aligned} \quad (\text{B1})$$

The single-path response function can therefore be integrated numerically using the initial condition $R_{i \setminus j}^{\text{sp}}(t, t') = 0$ for $t < t'$ from causality.

176 The numerical integration of the GECaM dynamics is performed using an Euler-Maruyama scheme with a time
 177 step Δ . Time is discretized in intervals of size Δ , i.e. $t = n\Delta$ with $n = 0, 1, \dots, T$, where T is the number of
 178 time steps. The trajectories $x(t)$ become T -dimensional time-vectors with elements $x^n = x(t = n\Delta)$, while the
 179 self-consistent fields $\mu(t)$, $R(t, t')$ and $C(t, t')$ become, respectively, T -dimensional vectors and $T \times T$ matrices, with
 180 elements $\mu^n = \mu(t = n\Delta)$, $R^{n,l} = R(t = n\Delta, t' = l\Delta)$ and $C^{n,l} = C(t = n\Delta, t' = l\Delta)$.

181 Both the effective cavity SDEs Eq. (4) and the effective marginal SDEs Eq. (7) are better integrated in log space.
 182 The discretized effective cavity SDE in log space reads

$$\log x_{i \setminus j}^{n+1} = \log x_{i \setminus j}^n + \Delta \mathcal{F}_{i \setminus j}^n \left(\mathbf{x}_{i \setminus j} \middle| \left\{ \mu_{k \setminus i}^n \right\}_{k \in \partial i \setminus j}, \left\{ \mathbf{R}_{k \setminus i} \right\}_{k \in \partial i \setminus j}, \zeta_{i \setminus j}^n \right) + \Delta \sqrt{2T} \mathcal{G} \left(x_{i \setminus j}^n \middle| \xi_{i \setminus j}^n \right) + \Delta \mathcal{H} \left(x_{i \setminus j}^n \middle| \lambda \right), \quad (\text{C1})$$

183 where

$$\mathcal{F}_{i \setminus j}^n \left(\mathbf{x}_{i \setminus j} \middle| \left\{ \mu_{k \setminus i}^n \right\}_{k \in \partial i \setminus j}, \left\{ \mathbf{R}_{k \setminus i} \right\}_{k \in \partial i \setminus j}, \zeta_{i \setminus j}^n \right) = 1 - x_{i \setminus j}^n + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}^n + \Delta \sum_{m=0}^{n-1} x_{i \setminus j}^m \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} R_{k \setminus i}^{n,m} + \zeta_{i \setminus j}^n, \quad (\text{C2})$$

$$\mathcal{G}(x | \xi) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ \exp[\log \xi - \frac{1}{2} \log x] & g(x) = x & \text{demographic} \\ \xi & g(x) = x^2 & \text{environmental,} \end{cases} \quad (\text{C3})$$

$$\mathcal{H}(x | \lambda) = \begin{cases} 0 & \lambda = 0 \\ \exp[\log \lambda - \log x] & \lambda > 0. \end{cases} \quad (\text{C4})$$

184 The discretized equation for the single path cavity response function reads

$$\begin{aligned} R_{i \setminus j}^{(\text{sp})n+1,l} &= R_{i \setminus j}^{(\text{sp})n,l} + \Delta R_{i \setminus j}^{(\text{sp})n,l} \left(1 - 2x_{i \setminus j}^n + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}^n + \zeta_{i \setminus j}^n + \sqrt{2T} \xi_{i \setminus j}^n \mathcal{I}(x_{i \setminus j}^n) \right) + x_{i \setminus j}^n \delta_{n,l} \\ &+ \Delta^2 R_{i \setminus j}^{(\text{sp})n,l} \sum_{m=0}^{n-1} x_{i \setminus j}^m \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} R_{k \setminus i}^{n,m} + \Delta^2 x_{i \setminus j}^n \sum_{m=l+1}^{n-1} R_{i \setminus j}^{(\text{sp})m,l} \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} R_{k \setminus i}^{n,m}, \end{aligned} \quad (\text{C5})$$

185 where

$$\mathcal{I}(x) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1/\sqrt{x} & g(x) = x & \text{demographic} \\ 1 & g(x) = x^2 & \text{environmental.} \end{cases} \quad (\text{C6})$$

186 The discretized equation for the single path marginal response function reads

$$\begin{aligned} R_i^{(\text{sp})n+1,l} &= R_i^{(\text{sp})n,l} + \Delta R_i^{(\text{sp})n,l} \left(1 - 2x_i^n + \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}^n + \zeta_i^n + \sqrt{2T} \xi_i^n \mathcal{I}(x_i^n) \right) + x_i^n \delta_{n,l} \\ &+ \Delta^2 R_i^{(\text{sp})n,l} \sum_{m=0}^{n-1} x_i^m \sum_{k \in \partial i} J_{ik} J_{ki} R_{k \setminus i}^{n,m} + \Delta^2 x_i^n \sum_{m=l+1}^{n-1} R_i^{(\text{sp})m,l} \sum_{k \in \partial i} J_{ik} J_{ki} R_{k \setminus i}^{n,m}. \end{aligned} \quad (\text{C7})$$

187 The discretized effective marginal SDE in log space reads

$$\log x_i^{n+1} = \log x_i^n + \Delta \mathcal{F}_i^n \left(\mathbf{x}_i \middle| \left\{ \mu_{k \setminus i}^n \right\}_{k \in \partial i}, \left\{ \mathbf{R}_{k \setminus i} \right\}_{k \in \partial i}, \zeta_i^n \right) + \Delta \sqrt{2T} \mathcal{G} \left(x_i^n \middle| \xi_i^n \right) + \Delta \mathcal{H} \left(x_i^n \middle| \lambda \right), \quad (\text{C8})$$

188 where

$$\mathcal{F}_i^n \left(\mathbf{x}_i \middle| \left\{ \mu_{k \setminus i}^n \right\}_{k \in \partial i}, \left\{ \mathbf{R}_{k \setminus i} \right\}_{k \in \partial i}, \zeta_i^n \right) = 1 - x_i^n + \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}^n + \Delta \sum_{m=0}^{n-1} x_i^m \sum_{k \in \partial i} J_{ik} J_{ki} R_{k \setminus i}^{n,m} + \zeta_i^n. \quad (\text{C9})$$

In order to avoid unnecessary computations, we can define the following node quantities:

$$\Sigma_{i,\mu}^n = \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}^n, \quad (\text{C10})$$

$$\Sigma_{i,R}^{n,l} = \sum_{k \in \partial i} J_{ik} J_{ki} R_{k \setminus i}^{n,l}, \quad (\text{C11})$$

$$\Sigma_{i,C}^{n,l} = \sum_{k \in \partial i} J_{ik}^2 C_{k \setminus i}^{n,l}, \quad (\text{C12})$$

$$(\text{C13})$$

which will be computed once at the beginning of the integration. The cavity contributions can therefore be computed by subtracting the contribution of the node j from the sum over the neighbors of node i , thus avoiding the need to compute the sum over all neighbors at each time step.

The resulting algorithm proceeds as follows.

1. Propose initial guesses for the cavity mean abundances $\{\mu_{i \setminus j}\}_{(i,j) \in E}$, the cavity correlations $\{C_{i \setminus j}\}_{(i,j) \in E}$, and the cavity responses $\{R_{i \setminus j}\}_{(i,j) \in E}$ for each edge of the graph.

2. For each node i of the graph:

- (a) Compute the node quantities $\Sigma_{i,\mu}^n$, $\Sigma_{i,R}^{n,l}$ and $\Sigma_{i,C}^{n,l}$ using Eqs. (C10), (C11) and (C12).

- (b) For each edge (i, j) of the graph:

- i. Sample N_{traj} Gaussian paths $\{(\zeta_{i \setminus j}^{(s)}, \xi_{i \setminus j}^{(s)})\}_{s=1}^{N_{\text{traj}}}$, where $\zeta_{i \setminus j}$ and $\xi_{i \setminus j}$ have, respectively, covariance matrices $C_{i \setminus j}^{(\zeta)n,l} = \Sigma_{i,C}^{n,l} - J_{ik}^2 C_{k \setminus i}^{n,l}$ and $C_{i \setminus j}^{(\xi)n,l} = \delta_{n,l}$. This can be done by diagonalizing the covariance matrix $C_{i \setminus j}^{(\zeta)}$ and using the proper basis to sample independent gaussian components.

- ii. For each path $(\zeta_{i \setminus j}^{(s)}, \xi_{i \setminus j}^{(s)})$, numerically integrate the effective cavity equation with an initial condition sampled according to $p_0(x)$ using the Euler-Maruyama scheme in log space, i.e. Eq. (C1). This will yield N_{traj} trajectories $\{x_{i \setminus j}^{(s)}\}_{s=1}^{N_{\text{traj}}}$.

- iii. For each path $(\zeta_{i \setminus j}^{(s)}, \xi_{i \setminus j}^{(s)})$, compute the single-path response function using Eq. (C5). This will yield N_{traj} single path response matrices $\{R_{i \setminus j}^{(s)}\}_{s=1}^{N_{\text{traj}}}$.

- iv. Compute the new values of the cavity self-consistent fields as averages over the paths:

$$\begin{aligned} \mu_{i \setminus j}^{(\text{new})n} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} x_{i \setminus j}^{(s)n}, \\ C_{i \setminus j}^{(\text{new})n,l} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} x_{i \setminus j}^{(s)n} x_{i \setminus j}^{(s)l} - \mu_{i \setminus j}^n \mu_{i \setminus j}^l, \\ R_{i \setminus j}^{(\text{new})n,l} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} R_{i \setminus j,k}^{(s)n,l}. \end{aligned}$$

3. Check for convergence of the cavity self-consistent fields.

$$\max_{(i,j)} \|C_{i \setminus j}^{\text{updated}} - C_{i \setminus j}^{\text{new}}\| < \epsilon, \quad (\text{C14})$$

If not converged, go to step 2.

4. For each node i of the graph:

- (a) Compute the node quantities $\Sigma_{i,\mu}^n$, $\Sigma_{i,R}^{n,l}$ and $\Sigma_{i,C}^{n,l}$ using Eqs. (C10), (C11) and (C12).

- (b) Sample N_{traj} Gaussian paths $\{(\zeta_i^{(s)}, \xi_i^{(s)})\}_{s=1}^{N_{\text{traj}}}$, where ζ_i and ξ_i have, respectively, covariance matrices $C_i^{(\zeta)n,l} = \Sigma_{i,C}^{n,l}$ and $C_i^{(\xi)n,l} = \delta_{n,l}$. This can be done by diagonalizing the covariance matrix $C_i^{(\zeta)}$ and using the proper basis to sample independent gaussian components.

- (c) For each path $(\zeta_i^{(s)}, \xi_i^{(s)})$, numerically integrate the effective marginal equation with an initial condition sampled according to $p_0(x)$ using the Euler-Maruyama scheme in log space, i.e. Eq. (C8). This will yield N_{traj} trajectories $\{x_i^{(s)}\}_{s=1}^{N_{\text{traj}}}$.
- (d) For each path $(\zeta_i^{(s)}, \xi_i^{(s)})$, compute the single-path response function using Eq. (C5). This will yield N_{traj} single path response matrices $\{R_i^{(s)}\}_{s=1}^{N_{\text{traj}}}$.
- (e) Compute the marginal self-consistent fields as averages over the paths:

$$\begin{aligned}\mu_i^n &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} x_i^{(s)n}, \\ C_i^{n,l} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} x_i^{(s)n} x_i^{(s)l} - \mu_i^n \mu_i^l, \\ R_i^{n,l} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} R_{i,k}^{(s)n,l}.\end{aligned}$$

Appendix D: Gaussian closure

The effective process Eq. (7) is a stochastic differential equation that can be solved numerically. However, it is often useful to obtain analytical closed form equations for the fields μ , C and R . For simplicity, we rewrite the effective cavity process in the form

$$\frac{d}{dt} x_{i \setminus j}(t) = x_{i \setminus j}(t) \left[1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i \setminus j} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_{i \setminus j}(t') \right] + \eta_{i \setminus j}(t) + \lambda, \quad (\text{D1})$$

where we have defined the noise term $\eta_{i \setminus j}(t)$ as a Gaussian process with vanishing mean and covariance

$$\langle \eta_{i \setminus j}(t) \eta_{i \setminus j}(t') \rangle = x_{i \setminus j}(t) \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t') x_{i \setminus j}(t') + 2Tg(x_{i \setminus j}(t)) \delta(t - t'). \quad (\text{D2})$$

The approximated cavity message can be written as

$$\begin{aligned}c_{i \setminus j}[x_{i \setminus j}, \hat{x}_{i \setminus j}] &= \left\langle \exp \int dt \left\{ -i \hat{x}_{i \setminus j}(t) \left[\frac{\dot{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} - \left(1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) \right. \right. \right. \right. \\ &\quad \left. \left. \left. + \sum_{k \in \partial i \setminus j} \int_0^t ds J_{ik} J_{ki} R_{k \setminus i}(t, s) x_{i \setminus j}(s) + \frac{\eta_{i \setminus j}(t)}{x_{i \setminus j}(t)} + \frac{\lambda}{x_{i \setminus j}(t)} \right) \right] \right\} \right\rangle\end{aligned} \quad (\text{D3a})$$

$$\begin{aligned}&= \exp \left\{ - \int dt \frac{i \hat{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} \left[\dot{x}_{i \setminus j}(t) - x_{i \setminus j}(t) \left(1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) \right. \right. \right. \\ &\quad \left. \left. \left. + \sum_{k \in \partial i \setminus j} \int_0^t ds J_{ik} J_{ki} R_{k \setminus i}(t, s) x_{i \setminus j}(s) \right) - \lambda \right] \right\} \left\langle \exp \int dt \frac{i \hat{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} \eta_{i \setminus j}(t) \right\rangle\end{aligned} \quad (\text{D3b})$$

$$\begin{aligned}&= \exp \left\{ - \int dt \frac{i \hat{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} \left[\dot{x}_{i \setminus j}(t) - x_{i \setminus j}(t) \left(1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) \right. \right. \right. \\ &\quad \left. \left. \left. + \sum_{k \in \partial i \setminus j} \int_0^t ds J_{ik} J_{ki} R_{k \setminus i}(t, s) x_{i \setminus j}(s) \right) - \lambda \right] \right\} \\ &\quad + \frac{1}{2} \int dt dt' \frac{i \hat{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} \left(2Tg(x_{i \setminus j}(t)) \delta(t - t') + x_{i \setminus j}(t) \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t') x_{i \setminus j}(t') \right) \frac{i \hat{x}_{i \setminus j}(t')}{x_{i \setminus j}(t')} \Bigg\}. \quad (\text{D3c})\end{aligned}$$

227 By averaging Eq. (4) over the noise realizations, we obtain the following self-consistent equations for the cavity
228 mean abundances

$$\begin{aligned} \frac{d}{dt}\mu_{i\setminus j}(t) &= \langle \dot{x}_{i\setminus j}(t) \rangle = \mu_{i\setminus j}(t) - C_{i\setminus j}(t, t) - \mu_{i\setminus j}^2(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) \mu_{i\setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_{i\setminus j}(t, s) + \mu_{i\setminus j}(t) \mu_{i\setminus j}(s)) + \langle \eta_{i\setminus j}(t) \rangle + \lambda \end{aligned} \quad (\text{D4a})$$

$$\begin{aligned} &= \mu_{i\setminus j}(t) - C_{i\setminus j}(t, t) - \mu_{i\setminus j}^2(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) \mu_{i\setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_{i\setminus j}(t, s) + \mu_{i\setminus j}(t) \mu_{i\setminus j}(s)) + \lambda. \end{aligned} \quad (\text{D4b})$$

229 Similarly, we can obtain the self-consistent equations for the disconnected cavity correlations

$$\begin{aligned} \frac{\partial}{\partial t} C_{i\setminus j}^{\text{dc}}(t, t') &= \langle \dot{x}_{i\setminus j}(t) x_{i\setminus j}(t') \rangle = C_{i\setminus j}^{\text{dc}}(t, t') - \langle x_{i\setminus j}^2(t) x_{i\setminus j}(t') \rangle + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) C_{i\setminus j}^{\text{dc}}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) x_{i\setminus j}(s) x_{i\setminus j}(t') \rangle + \langle \eta_{i\setminus j}(t) x_{i\setminus j}(t') \rangle + \lambda \mu_{i\setminus j}(t') \end{aligned} \quad (\text{D5a})$$

$$\begin{aligned} &= C_{i\setminus j}^{\text{dc}}(t, t') - \langle x_{i\setminus j}^2(t) x_{i\setminus j}(t') \rangle + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) C_{i\setminus j}^{\text{dc}}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) x_{i\setminus j}(s) x_{i\setminus j}(t') \rangle + 2T \langle \frac{g(x_{i\setminus j}(t))}{x_{i\setminus j}(t)} \dot{x}_{i\setminus j}(t) x_{i\setminus j}(t') \rangle \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}^2 \int ds C_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) \dot{x}_{i\setminus j}(s) x_{i\setminus j}(t') \rangle + \lambda \mu_{i\setminus j}(t'). \end{aligned} \quad (\text{D5b})$$

230 The self-consistent equations for the cavity response functions can be obtained as

$$\begin{aligned} \frac{\partial}{\partial t} R_{i\setminus j}(t, t') &= \langle \dot{x}_{i\setminus j}(t) \dot{x}_{i\setminus j}(t') \rangle = R_{i\setminus j}(t, t') - \langle x_{i\setminus j}^2(t) \dot{x}_{i\setminus j}(t') \rangle + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) R_{i\setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int ds R_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) x_{i\setminus j}(s) \dot{x}_{i\setminus j}(t') \rangle + \langle \eta_{i\setminus j}(t) \dot{x}_{i\setminus j}(t') \rangle. \end{aligned} \quad (\text{D6a})$$

$$\begin{aligned} &= R_{i\setminus j}(t, t') - \langle x_{i\setminus j}^2(t) \dot{x}_{i\setminus j}(t') \rangle + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) R_{i\setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int ds R_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) x_{i\setminus j}(s) \dot{x}_{i\setminus j}(t') \rangle + 2T \langle \frac{g(x_{i\setminus j}(t))}{x_{i\setminus j}(t)} \dot{x}_{i\setminus j}(t) \dot{x}_{i\setminus j}(t') \rangle \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}^2 \int ds C_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) \dot{x}_{i\setminus j}(s) \dot{x}_{i\setminus j}(t') \rangle. \end{aligned} \quad (\text{D6b})$$

231 The equations for the correlations and responses are not closed, as they depend on three-, four- and five-times averages.

232 We can close the equations by approximating those averages with sums of products of two-times averages, i.e. we can

233 use the Gaussian closure approximation. Specifically, introducing the zero-mean degrees of freedom $\delta x(t) = x(t) - \mu(t)$,

234 we can approximate as:

$$\begin{aligned} \langle x^2(t) x(t') \rangle &= \langle (\delta x(t) + \mu(t))^2 (\delta x(t') + \mu(t')) \rangle \approx C(t, t) \mu(t') + \mu^2(t) \mu(t') + 2\mu(t) C(t, t'), \\ \langle x(t) x(s) x(t') \rangle &= \langle (\delta x(t) + \mu(t)) (\delta x(s) + \mu(s)) (\delta x(t') + \mu(t')) \rangle \\ &\approx C(t, s) \mu(t') + C(t, t') \mu(s) + C(s, t') \mu(t) + \mu(t) \mu(s) \mu(t'), \\ \langle x(t) \dot{x}(s) x(t') \rangle &= \langle (\delta x(t) + \mu(t)) \dot{x}(s) (\delta x(t') + \mu(t')) \rangle \approx \mu(t) R(t', s) + R(t, s) \mu(t'), \\ \langle x^2(t) \dot{x}(t') \rangle &= \langle (\delta x(t) + \mu(t))^2 \dot{x}(t') \rangle \approx 2\mu(t) R(t, t'), \\ \langle x(t) x(s) \dot{x}(t') \rangle &= \langle (\delta x(t) + \mu(t)) (\delta x(s) + \mu(s)) \dot{x}(t') \rangle \approx R(t, t') \mu(s) + \mu(t) R(s, t'), \\ \langle x(t) \dot{x}(s) \dot{x}(t') \rangle &= \langle (\delta x(t) + \mu(t)) \dot{x}(s) \dot{x}(t') \rangle \approx 0. \end{aligned}$$

235 The terms involving the function $g(x)$ depend on the specific form of the function.

236 • **Demographic:** $g(x) = x$

$$\begin{aligned}\langle \frac{g(x(t))}{x(t)} \hat{x}(t) x(t') \rangle &= \langle \hat{x}(t) x(t') \rangle = R(t', t), \\ \langle \frac{g(x(t))}{x(t)} \hat{x}(t) \hat{x}(t') \rangle &= \langle \hat{x}(t) \hat{x}(t') \rangle = 0.\end{aligned}$$

237 • **Environmental:** $g(x) = x^2$

$$\begin{aligned}\langle \frac{g(x(t))}{x(t)} \hat{x}(t) x(t') \rangle &= \langle (\delta x(t) + \mu(t)) \hat{x}(t) (\delta x(t') + \mu(t')) \rangle \approx \mu(t) R(t', t), \\ \langle \frac{g(x(t))}{x(t)} \hat{x}(t) \hat{x}(t') \rangle &= \langle (\delta x(t) + \mu(t)) \hat{x}(t) \hat{x}(t') \rangle \approx 0.\end{aligned}$$

238 The resulting closed equations for the cavity mean abundances, correlations and responses read

$$\begin{aligned}\frac{d}{dt} \mu_{i \setminus j}(t) &= \mu_{i \setminus j}(t) - C_{i \setminus j}(t, t) - \mu_{i \setminus j}^2(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) \mu_{i \setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k \setminus i}(t, s) (C_{i \setminus j}(t, s) + \mu_{i \setminus j}(t) \mu_{i \setminus j}(s)) + \lambda,\end{aligned}\quad (D7)$$

$$\begin{aligned}\frac{\partial}{\partial t} C_{i \setminus j}^{\text{dc}}(t, t') &= C_{i \setminus j}^{\text{dc}}(t, t') (1 - 2\mu_{i \setminus j}(t)) - C_{i \setminus j}^{\text{dc}}(t, t) \mu_{i \setminus j}(t') + 2\mu_{i \setminus j}^2(t) \mu_{i \setminus j}(t') + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) C_{i \setminus j}^{\text{dc}}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k \setminus i}(t, s) (C_{i \setminus j}^{\text{dc}}(t, s) \mu_{i \setminus j}(t') + \mu_{i \setminus j}(t) C_{i \setminus j}^{\text{dc}}(s, t') + C_{i \setminus j}^{\text{dc}}(t, t') \mu_{i \setminus j}(s)) \\ &- 2 \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k \setminus i}(t, s) \mu_{i \setminus j}(t) \mu_{i \setminus j}(s) \mu_{i \setminus j}(t') + 2T \mathcal{L}_{i \setminus j}(t) R_{i \setminus j}(t', t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}^2 \int ds C_{k \setminus i}(t, s) (\mu_{i \setminus j}(t) R_{i \setminus j}(t', s) + R_{i \setminus j}(t, s) \mu_{i \setminus j}(t')) + \lambda \mu_{i \setminus j}(t').\end{aligned}\quad (D8)$$

$$\begin{aligned}\frac{\partial}{\partial t} R_{i \setminus j}(t, t') &= R_{i \setminus j}(t, t') (1 - 2\mu_{i \setminus j}(t)) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) R_{i \setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int ds R_{k \setminus i}(t, s) (R_{i \setminus j}(t, t') \mu_{i \setminus j}(s) + \mu_{i \setminus j}(t) R_{i \setminus j}(s, t')).\end{aligned}\quad (D9)$$

239 where we have introduced the function

$$\mathcal{L}_{i \setminus j}(t) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1 & g(x) = x & \text{demographic} \\ \mu_{i \setminus j}(t) & g(x) = x^2 & \text{environmental.} \end{cases}\quad (D10)$$

240 We can finally obtain the self-consistent equation for the connected cavity correlations by subtracting $\dot{\mu}_{i \setminus j}(t) \mu_{i \setminus j}(t)$
241 from the self-consistent equation for the disconnected cavity correlations:

$$\begin{aligned}\frac{\partial}{\partial t} C_{i \setminus j}(t, t') &= (1 - 2\mu_{i \setminus j}(t)) C_{i \setminus j}(t, t') + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) C_{i \setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k \setminus i}(t, s) (\mu_{i \setminus j}(t) C_{i \setminus j}(s, t') + C_{i \setminus j}(t, t') \mu_{i \setminus j}(s)) \\ &+ 2T \mathcal{L}_{i \setminus j}(t) R_{i \setminus j}(t', t) + \sum_{k \in \partial i \setminus j} J_{ik}^2 \int_0^{\max(t, t')} ds C_{k \setminus i}(t, s) (\mu_{i \setminus j}(t) R_{i \setminus j}(t', s) + R_{i \setminus j}(t, s) \mu_{i \setminus j}(t')).\end{aligned}\quad (D11)$$

242 Appendix E: Average over disorder vs small coupling expansion

243 When the average over disordered interactions is taken into account, one has to deal with an integral of the form

$$\overline{\int \mathcal{D}y \mathcal{D}\hat{y} \mathcal{D}w \mathcal{D}\hat{w} c[y, \hat{y}, w, \hat{w}] \exp \left[-i \int dt \left(J\hat{w}(t)x(t) + J'\hat{f}(t)y(t) \right) \right]}, \quad (\text{E1})$$

244 where $\overline{\cdots}$ denotes the average over disordered couplings (J, J') , i.e. $\overline{\cdots} = \int dJ dJ' p(J, J') \cdots$, where $p(J, J')$ is a
 245 bivariate Gaussian probability distribution function with averages $\overline{J} = \overline{J'} = m/K$, variances $\overline{(J - \overline{J})^2} = \overline{(J' - \overline{J'})^2} =$
 246 σ^2/K and correlation $\overline{(J - \overline{J})(J' - \overline{J'})} = \gamma\sigma^2/K$.

247 Here we demonstrate that when dealing with a large connectivity expansion the average over disorder can be
 248 exchanged with the small-coupling expansion, provided one only retains terms of order $1/K$ in the expansion.

249 We use the following shorthand notation for the sake of shortness: $\mathcal{D}c = \mathcal{D}y \mathcal{D}\hat{y} \mathcal{D}w \mathcal{D}\hat{w} c[y, \hat{y}, w, \hat{w}]$, $a \cdot b =$
 250 $\int dt a(t)b(t)$, with a and b being $x, y, \hat{f}$ and $\hat{w}$.

251 Average over disorder first

252 Since the average over disorder involves a Gaussian integral, it can be easily computed as follows

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w} \cdot x + J'\hat{f} \cdot y)}} = \int \mathcal{D}c e^{-i(J\hat{w} \cdot x + J'\hat{f} \cdot y)} \quad (\text{E2})$$

$$= \int \mathcal{D}c \exp \left[-\frac{m}{K} (\hat{w} \cdot x + \hat{f} \cdot y) - \frac{\sigma^2}{2K} ((\hat{w} \cdot x)^2 + (\hat{f} \cdot y)^2 - 2\gamma(\hat{w} \cdot x)(\hat{f} \cdot y)) \right]. \quad (\text{E3})$$

253 Expanding at first order in $1/K$, i.e. large connectivity expansion

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w} \cdot x + J'\hat{f} \cdot y)}} = \int \mathcal{D}c \left[1 - \frac{m}{K} (\hat{w} \cdot x + \hat{f} \cdot y) - \frac{\sigma^2}{2K} ((\hat{w} \cdot x)^2 + (\hat{f} \cdot y)^2 - 2\gamma(\hat{w} \cdot x)(\hat{f} \cdot y)) \right] \quad (\text{E4})$$

$$\approx 1 - \frac{m}{K} \int dt \left(\langle \hat{w}(t) \rangle x(t) + \langle \hat{f}(t) \rangle y(t) \right) + \gamma \frac{\sigma^2}{K} \int dt dt' \langle y(t) \hat{w}(t') \rangle \langle \hat{f}(t') x(t') \rangle$$

$$- \frac{\sigma^2}{2K} \int dt dt' \left(\langle \hat{w}(t) \hat{w}(t') \rangle x(t) x(t') + \langle \hat{f}(t) \hat{f}(t') \rangle y(t) y(t') \right) \quad (\text{E5})$$

$$= 1 - \frac{m}{K} \int dt \langle \hat{f}(t) \rangle \mu(t) + \gamma \frac{\sigma^2}{K} \int dt dt' \langle \hat{f}(t) R(t, t') x(t') \rangle - \frac{\sigma^2}{2K} \int dt dt' \hat{f}(t) C^{\text{dc}}(t, t') \hat{f}(t'), \quad (\text{E6})$$

254 where we have defined the average over the measure $c[y, \hat{y}, w, \hat{w}]$ as $\langle \cdots \rangle$. We have introduced the average $\mu(t) = \langle y(t) \rangle$,
 255 the (disconnected) correlation function $C^{\text{dc}}(t, t') = \langle y(t) y(t') \rangle$ and the response function $R(t, t') = \langle y(t) \hat{w}(t') \rangle$. The
 256 averages of hatted quantities only vanish due to causality constraints. By re-exponentiating we get the following
 257 approximate expression

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w} \cdot x + J'\hat{f} \cdot y)}} \approx \exp \left\{ -\frac{m}{K} \int dt \langle \hat{f}(t) \rangle \mu(t) + \gamma \frac{\sigma^2}{K} \int dt dt' \langle \hat{f}(t) R(t, t') x(t') \rangle \right. \\ \left. - \frac{\sigma^2}{2K} \int dt dt' \hat{f}(t) C^{\text{dc}}(t, t') \hat{f}(t') \right\}. \quad (\text{E7})$$

259 We expand the exponential at second order in J , i.e. small coupling expansion,

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx \overline{\int \mathcal{D}c \left[1 - \left(J\hat{w}\cdot x + J'\hat{f}\cdot y \right) + \frac{1}{2} \left(J\hat{w}\cdot x + J'\hat{f}\cdot y \right)^2 \right]} \quad (\text{E8})$$

$$= 1 - \int dt \left(J\langle \hat{w}(t) \rangle x(t) + J'\langle \hat{f}(t) \rangle y(t) \right) + \frac{1}{2} \int dt dt' \left(J^2 \langle \hat{w}(t) \hat{w}(t') \rangle x(t) x(t') \right. \\ \left. + J'^2 \langle \hat{f}(t) \hat{f}(t') \rangle y(t) y(t') + 2JJ' \langle y(t) \hat{w}(t') \rangle \langle \hat{f}(t) x(t') \rangle \right) \quad (\text{E9})$$

$$= 1 - \int dt J'\langle \hat{f}(t) \rangle \mu(t) + \frac{1}{2} \int dt dt' \left(J'^2 \langle \hat{f}(t) \rangle C^{\text{dc}}(t, t') \langle \hat{f}(t') \rangle + 2JJ' \langle \hat{f}(t) \rangle R(t, t') x(t') \right). \quad (\text{E10})$$

 260 By re-exponentiating, we have to carefully consider the square of first order terms, i.e. $1 - \langle x \rangle + \langle x^2 \rangle / 2 \approx \exp(\langle x \rangle +$
 261 $\langle x^2 \rangle / 2 - \langle x \rangle^2 / 2)$ when $x \approx 0$.

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx \overline{\exp \left\{ - \int dt J'\langle \hat{f}(t) \rangle \mu(t) + \frac{1}{2} \int dt dt' J'^2 \langle \hat{f}(t) \rangle C(t, t') \langle \hat{f}(t') \rangle \right.} \\ \left. + \int dt dt' JJ' \langle \hat{f}(t) \rangle R(t, t') x(t') \right\}}, \quad (\text{E11})$$

 262 where we have introduced the (connected) correlation function $C(t, t') = C^{\text{dc}}(t, t') - \mu(t)\mu(t')$. We defined, for the
 263 sake of shortness, the terms $A = \int dt \langle \hat{f}(t) \rangle \mu(t)$, $B = \int dt dt' \langle \hat{f}(t) \rangle C(t, t') \langle \hat{f}(t') \rangle$, and $C = \int dt dt' \langle \hat{f}(t) \rangle R(t, t') x(t')$. By
 264 Gaussian integration, we obtain

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx \exp \left(-AJ' + \frac{B}{2}J'^2 + CJJ' \right) \quad (\text{E12})$$

$$= \frac{\exp \left[\frac{\sigma^2 (A^2 K^2 + 2AC(\gamma-1)Km - 2C^2(\gamma-1)m^2) + Km(m(B+2C) - 2AK)}{2K(-K\sigma^2(B+2C\gamma) + C^2(\gamma^2-1)\sigma^4 + K^2)} \right]}{\sqrt{1 + \frac{C^2(\gamma^2-1)\sigma^4 - K\sigma^2(B+2C\gamma)}{K^2}}}. \quad (\text{E13})$$

 265 Retaining only the first order terms in $1/K$, i.e. large connectivity expansion

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx 1 - \frac{m}{K}A + \frac{\sigma^2}{2K} (A^2 + B + 2\gamma C) \quad (\text{E14})$$

$$\approx \exp \left[-\frac{m}{K}A + \frac{\sigma^2}{2K} (A^2 + B + 2\gamma C) \right] \quad (\text{E15})$$

$$= \exp \left\{ -\frac{m}{K} \int dt \langle \hat{f}(t) \rangle \mu(t) + \frac{\sigma^2}{2K} \int dt dt' \langle \hat{f}(t) \rangle (\mu(t)\mu(t') + C(t, t')) \langle \hat{f}(t') \rangle \right. \\ \left. + \gamma \frac{\sigma^2}{K} \int dt dt' \langle \hat{f}(t) \rangle R(t, t') x(t') \right\} \quad (\text{E16})$$

$$= \exp \left\{ -\frac{m}{K} \int dt \langle \hat{f}(t) \rangle \mu(t) + \frac{\sigma^2}{2K} \int dt dt' \langle \hat{f}(t) \rangle C^{\text{dc}}(t, t') \langle \hat{f}(t') \rangle \right. \\ \left. + \gamma \frac{\sigma^2}{K} \int dt dt' \langle \hat{f}(t) \rangle R(t, t') x(t') \right\}, \quad (\text{E17})$$

266 which is the same approximate result obtained by taking first the average over disorder. We note that the same result
 267 is obtained if we take the average over disorder before re-exponentiating, i.e in Eq. (E10),

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx 1 - \int dt \overline{J'} i\hat{f}(t) \mu(t) + \frac{1}{2} \int dt dt' \left(\overline{J'^2} i\hat{f}(t) C^{\text{dc}}(t, t') i\hat{f}(t') + 2\overline{J'J'} i\hat{f}(t) R(t, t') x(t') \right) \quad (\text{E18})$$

$$\approx 1 - \int dt \frac{m}{K} i\hat{f}(t) \mu(t) + \frac{1}{2} \int dt dt' \left(\frac{\sigma^2}{K} i\hat{f}(t) C^{\text{dc}}(t, t') i\hat{f}(t') + 2\gamma \frac{\sigma^2}{K} i\hat{f}(t) R(t, t') x(t') \right) \quad (\text{E19})$$

$$\approx \exp \left\{ -\frac{m}{K} \int dt i\hat{f}(t) \mu(t) + \frac{\sigma^2}{2K} \int dt dt' i\hat{f}(t) C^{\text{dc}}(t, t') i\hat{f}(t') \right. \\ \left. + \gamma \frac{\sigma^2}{K} \int dt dt' i\hat{f}(t) R(t, t') x(t') \right\}. \quad (\text{E20})$$